PROJECT PHASE 2

DATABASE DESIGN

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1.Introduction

This document details the database design phase for the Royal Fields project. It includes the business rules derived from the company's operations, an Entity Relationship (ER) diagram, and a logical data model. This design will serve as the foundation for the physical database implementation in Phase 3.

2.Business Rules

The following business rules must be adhered to when the new database is being designed for Royal Fields, ensuring alignment with the company operations and the relationships defined in the ERD:

Members Subscribe to Receive Weekly or Bi-Weekly Boxes of Seasonal Produce

- **Rule 1.1**: Members must subscribe to receive produce boxes, and each subscription must be tied to a single member.
  A member must have at least one subscription to participate in the service, and each subscription must belong to exactly one member. This ensures that Royal Fields can track which member is associated with each subscription for delivery and payment purposes.
- **Rule 1.2**: Each subscription must specify a weekly or bi-weekly delivery frequency for the member's produce boxes.
  The subscription must define the delivery frequency as either weekly or bi-weekly to align with member expectations and operational planning. This ensures that deliveries are scheduled consistently with the member's chosen frequency.

Farmers Supply Produce to Royal Fields, Adhering to Quality and Cultural Relevance Standards

- **Rule 2.1**: Farmers must supply at least one batch of produce to participate, and each batch must be recorded in the inventory.
  To be active participants, farmers must provide at least one batch of produce, which is recorded in the inventory. This ensures a consistent supply of produce for member deliveries and tracks farmer contributions.
- **Rule 2.2**: Produce supplied by farmers must meet quality and cultural relevance standards, including traditional African vegetables and fruits.
  All produce must adhere to Royal Fields' quality and cultural relevance standards, ensuring it includes traditional African vegetables and fruits that align with the dietary preferences of the Limpopo community. This maintains the cultural relevance and quality expected by members.

Royal Fields Staff Coordinate Farmer Relationships, Manage Subscriptions, Plan Delivery Logistics, and Handle Financial Transactions

- **Rule 3.1**: Royal Fields staff must manage subscriptions, ensuring each member has at least one active subscription.
  Staff manage subscriptions by ensuring that each member has at least one active subscription, facilitating the delivery of produce boxes and maintaining member engagement.

- **Rule 3.2**: Royal Fields staff must plan delivery logistics by assigning each delivery to a strategic delivery hub and route, but a route may have no deliveries scheduled.
  Staff plan delivery logistics by assigning deliveries to strategic hubs and routes, but a route may have zero or many deliveries, as some routes might not have deliveries scheduled yet (e.g., new routes). This ensures efficient distribution while allowing operational flexibility.

- **Rule 3.3**: Royal Fields staff must handle financial transactions, ensuring members and farmers can have at most one payment record, but payments are optional.
  Staff handle financial transactions, allowing a member or farmer to have zero or one payment record, as they might not have made or received a payment yet (e.g., on a trial or new supplier). This ensures flexibility while tracking financial transactions.

Produce Boxes Contain a Variety of Seasonal Produce, Including Traditional African Vegetables and Fruits

- **Rule 4.1**: Each produce box must contain exactly one type of produce for delivery, ensuring a variety of seasonal produce over time.
  For simplicity, each delivery contains exactly one box content record (one type of produce per delivery), but over multiple deliveries, members receive a variety of seasonal produce, including traditional African vegetables and fruits. This ensures diversity over time while simplifying delivery tracking.

- **Rule 4.2**: Produce must be included in at least one delivery box to be considered active in the system.
  For a produce type to be active, it must be included in at least one delivery box, ensuring that only relevant seasonal produce, including traditional African items, is maintained in the system for member deliveries.

Strategic Delivery Hubs Are Established Across Limpopo to Facilitate Efficient Distribution

- **Rule 5.1**: Deliveries must be scheduled for members, but a member may have no deliveries yet.
  Deliveries are scheduled for members through strategic hubs, but a member may have zero or many deliveries, as they might not have received any yet (e.g., new member). This ensures flexibility for new members while supporting regular deliveries.

- **Rule 5.2**: Each delivery must have a unique status to track its progress through the distribution hubs.
  Each delivery must have exactly one status (e.g., "Scheduled," "Delivered") to monitor its

progress through the strategic hubs, ensuring efficient logistics and transparency for members.

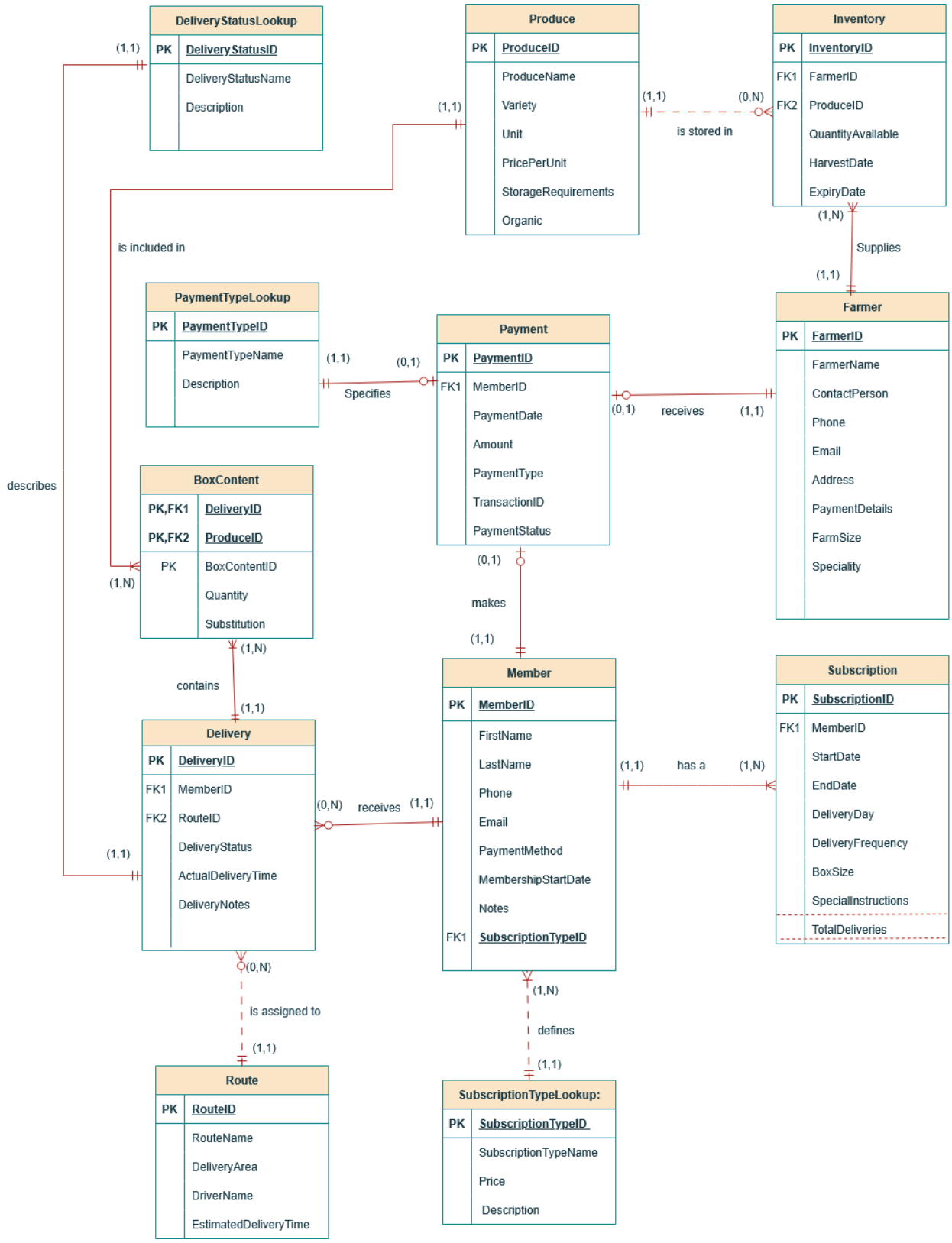
Flexible Payment Options Are Implemented, Considering the Diverse Economic Circumstances of the Region

- **Rule 6.1**: Payment types must be predefined, and each payment can use at most one payment type, but some payment types may not be used.
  Flexible payment options are implemented, so each payment must use one predefined payment type (e.g., "Credit Card," "Bank Transfer"), but a payment type can be associated with zero or one payment, as some types might not be used yet. This ensures payment flexibility while standardizing options.

General Rules Supporting Multiple Operations

Each subscription type must be used by at least one member, ensuring all types are active. Members subscribe to receive produce boxes, and their subscription type must be one of the predefined options. Each subscription type must be used by at least one member to ensure all types are active, supporting the management of subscriptions by Royal Fields staff.

3.Entity Relationship Diagram



4.Notes on the ER diagram

Bridge Entities

- **BoxContent**: A bridge entity between Delivery and Produce with its own unique ID (BOX_CONTENT_ID), making it a strong entity.

Strong Relationships

- Member and Subscription
- Member and Delivery
- Member and Payment
- Farmer and Inventory
- Farmer and Payment
- Produce and BoxContent
- Delivery and BoxContent
- DeliveryStatusLookup and Delivery
- PaymentTypeLookup and Payment

Weak Relationships

- Produce and Inventory
- Route and Delivery
- SubscriptionTypeLookup and Member

Mandatory Relationships

- Member and Subscription
- Delivery and BoxContent
- DeliveryStatusLookup and Delivery
- Farmer and Inventory
- Produce and BoxContent
- Delivery and Member
- Delivery and Route
- Payment and Member
- Payment and Farmer
- Payment and PaymentTypeLookup

- Inventory and Produce

Derived Attributes

- TOTAL_DELIVERIES in Subscription

Composite Attributes

- Address in Farmer

5.Logical Data Model

Mapping the Conceptual Model to the Logical Model

1. **Strong Entities Were Mapped**:
    - All entities in the ERD are strong entities: DeliveryStatusLookup, Produce, Inventory, Farmer, PaymentTypeLookup, Payment, Member, Subscription, SubscriptionTypeLookup, Delivery, Route, and BoxContent. There are no weak entities, as BoxContent uses a surrogate key (BOX_CONTENT_ID) to ensure uniqueness.
2. **Binary Relationships Were Mapped**:
    - All relationships are binary (between two entities), such as Member → Subscription, Delivery → BoxContent, Farmer → Inventory, and Payment → PaymentTypeLookup. These are represented using foreign keys in the child entities.

Logical Model

Below is the logical model with relations for each entity.
Primary keys (PK) are underlined, and foreign keys (FK) are italicized. The model is 3NF.

- **DELIVERYSTATUSLOOKUP** (**DELIVERY_STATUS_ID** (PK), DeliveryStatusName, Description)
- **PRODUCE** (**PRODUCE_ID**(PK), ProduceName, Variety, Unit, PricePerUnit, StorageRequirements, Organic)
- **INVENTORY** (**INVENTORY_ID** (PK), *FARMER_ID* (FK1), *PRODUCE_ID* (FK2), QuantityAvailable, HarvestDate, ExpiryDate)
- **FARMER** (**FARMER_ID**(PK), FarmerName, ContactPerson, Phone, Email, Street, City, Zip, PaymentDetails, FarmSize, Specialty)

- **PAYMENTTYPELOOKUP** (**PAYMENT_TYPE_ID**(PK), PaymentTypeName, Description)
- **PAYMENT** (**PAYMENT_ID** (PK), *MEMBER_ID* (FK1), *PAYMENT_TYPE_ID* (FK2), *FARMER_ID* (FK3), Amount, PaymentDate, TransactionID)
- **MEMBER** (**MEMBER_ID** (PK), FirstName, LastName, Phone, Email, PaymentMethod, MembershipStartDate, Notes, *SUBSCRIPTION_TYPE_ID* (FK1))
- **SUBSCRIPTION** (**SUBSCRIPTION_ID**(PK), *MEMBER_ID* (FK1), StartDate, EndDate, DeliveryDay, DeliveryFrequency, BoxSize, SpecialInstructions, TOTAL_DELIVERIES)
    - *TOTAL_DELIVERIES is a derived attribute (not stored, computed at query time).*
- **SUBSCRIPTIONTYPELOOKUP** (**SUBSCRIPTION_TYPE_ID**(PK), SubscriptionTypeName, Price, Description)
- **DELIVERY** (**DELIVERY_ID**(PK), *MEMBER_ID* (FK1), *ROUTE_ID* (FK2), *DELIVERY_STATUS_ID* (FK3), ActualDeliveryTime, DeliveryNotes)
- **ROUTE** (**ROUTE_ID**(PK), RouteName, DeliveryArea, DriverName, EstimatedDeliveryTime)
- **BOXCONTENT** (**BOX_CONTENT_ID** (PK), *DELIVERY_ID* (FK1), *PRODUCE_ID* (FK2), Quantity)